SEMESTER – VI

MANAV RACHNA INTERNATIONAL INSTITUTE OF RESEARCH AND STUDIES
(Deemed to be University under section 3 of the UGC Act 1956)
NAAC 'A++' Grade University

PROJ-CS-600: PROJECT PHASE -I

Periods/week Credits
P: 2 1.0
Duration of Examination: 2 Hrs

Max. Marks : 50
Continuous Evaluation : 50

Pre-Requisite: NIL
Course Type: PROJ

Course Outcomes: Students will be able to-
PROJ-CS-600.1. Conceptualize real world situations related to systems development decisions, Originating from source requirements and goals.
PROJ-CS-600.2. Identify various computing tools.
PROJ-CS-600.3. Identify their area of interest and do extensive literature survey on the same.
PROJ-CS-600.4. Design models as a solution for particular problems.
PROJ-CS-600.5. Implement basic solutions in various platforms
PROJ-CS-600.6. Prepare a summarized report in the form of synopsis.

Text Books / Reference Books:
1. Harold Kerzner, 2013, Project Management: A Systems Approach to Planning, Scheduling, and Controlling; 11th edition, WILEY.
2. Adrienne Watt, 2008, Project management; BC Open Text book.

Software required/Weblinks:
Ieee.org
www.neptal.com

Distribution of Continuous Evaluation:

Presentation/Viva	40%
Report	20%
Class Work/ Performance	20%
Attendance	20%

Evaluation Tools:
Presentation/Implementation

COURSE ARTICULATION MATRIX:

CO Statement (PROJ-CS-600)	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
PROJ-CS-600.1	-	-	-	2	-	3	3	1	-	-	-	-	-	-	-

PROJ-CS-600.2	-	-	-	-	2	-	-	-	-	-	-	-	-	-	-
PROJ-CS-600.3	-	2	-	1	-	-	-	-	-	-	-	-	-	-	3
PROJ-CS-600.4	-	-	3	-	-	-	-	-	-	-	1	-	-	-	-
PROJ-CS-600.5	-	-	-	2	3	-	-	-	-	-	-	-	-	-	-
PROJ-CS-600.6	-	-	-	3	-	-	-	3	2	-	2	2	-	-	-

MANAV RACHNA INTERNATIONAL INSTITUTE OF RESEARCH AND STUDIES
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BHM-MC-009: QUANTITATIVE APTITUDE AND PERSONALITY DEVELOPMENT-III

Periods/week	Credits	Max. Marks	: 100
P :2	AP	Continuous Evaluation	: 50
Duration of Exam: 2 Hrs		End Sem Examination	: 50

Pre-Requisite: NIL
Course Type: HSMC

Course Outcomes: Students will be able to-
BHM-MC-009.1. Recognize problem based on Modern Mathematics and Algebra
BHM-MC-009.2. Solve basic to moderate level problems based on Mensuration and Geometry.
BHM-MC-009.3. Calculate solution to logical reasoning.
BHM-MC-009.4. Get proficient with resume building and will be able to draft effective cover letters.
BHM-MC-009.5. Participate effectively and confidently in a Group Discussion
BHM-MC-009.6. Manage interviews effectively.

PART – A

Unit 1: Modern Mathematics and Algebra
1.1 Permutation and Combination
1.1.1 Principal of counting and Basic formulas
1.1.2 Arrangements, Selection and Selection + Arrangement.
1.1.3 Linear/Circular arrangements, Digits and Alphabetic Problems and Applications.
1.2 Probability
1.2.1 Events and Sample Space, Basic Formulas.
1.2.2 Problems on Coins, Cards and Dices.
1.2.3 Conditional Probability, Bayes' Theorem and their Applications.
1.3 Algebra
1.3.1 Linear & Quadratic equations
1.3.2 Mathematical inequalities
1.3.4 Maximum & Minimum Values
1.3.3 Integral Solutions

Unit 2: Geometry and Mensuration
2.1 Geometry
2.1.1 Basic geometry & Theorems, Lines & Angles
2.1.2 Polygons, Triangle and Quadrilaterals
2.1.3 Circles
2.2 Mensuration I- Areas
2.2.1 Different types of Triangles and their area and perimeter.
2.2.2 Different types of Quadrilateral and their area and perimeter.
2.2.3 Circumference and Area of Circle, Area of Sector and length of Sector.
2.2.4 Mixed Figures and their Applications.
2.3 Mensuration II- Surface Areas and Volumes

2.3.1 Problems on Cubes & Cuboids, Cone, Cylinder and Sphere.
2.3.2 Prism and Pyramid.
2.3.3 Mixed Figures and their Applications.

Unit 3: Logical Reasoning
3.1 Linear Arrangement
3.2 Circular Arrangement
3.3 Puzzles

Part - B

Unit 4: Professional Writing
4.1. Profiling on Social Sites: LinkedIn, Facebook, Instagram
4.2. Cover Letter/Emails
4.3. Resume Writing

Unit 5: Group Discussions
5.1. Do's and Dont's of a Group Discussion
5.2. Roles played in a Group Discussion
5.3. Tips for Cracking a Group Discussion

Unit 6: Managing Interviews
6.1. Developing the employability mindset
6.2. Preparing for Self -Introduction
6.3. Researching the employer
6.4. Portfolio Management
6.5. Answering Questions in an Interview

Text Books/Reference Books:
1. Arun Sharma, 2017 Teach Your Self Quantitative Aptitude, 1st Edition, McGraw Hills Education.
2. R S Aggarwal,2017, A Modern Approach to Logical Reasoning, S Chand & Company Pvt Ltd.
3. Yana Parker & Beth Brown, The Damn Good resume Guide
4. Ceri Roderick & Stephan Lucks, Interview Answers

Instructions for paper setting: Fifty MCQ will be set in total. Twenty five MCQ will be set from Part A and twenty five MCQ will be set from Part B. All questions will be compulsory. Each question will be of 1 mark. There will be no negative marking. Calculator will not be allowed.

Distribution of Continuous Evaluation:

Sessional- I	30%
Sessional- II	30%
Assignment/Tutorial	20%
Class Work/ Performance	10%

Attendance	10%

Course Articulation Matrix :

CO Statement (BHM-MC-009)	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
BHM-MC-009.1	1	-	-	-	-	1	-	-	-	-	-	1	-	-	1
BHM-MC-009.2	1	-	-	-	-	1	-	-	-	-	-	1	-	-	1
BHM-MC-009.3	1	-	-	2	-	-	-	-	-	-	-	-	-	-	-
BHM-MC-009.4	-	-	-	-	-	-	-	1	-	3	-	1	-	-	-
BHM-MC-009.5	-	-	-	-	-	-	-	1	-	3	-	-	-	-	-
BHM-MC-009.6	1	-	-	-	-	1	-	-	-	-	-	1	-	-	1

Evaluation Tools:
Experiments in lab
File work/Class Performance
Viva (Question and answers in lab)
End Semester Practical Examination

COURSE ARTICULATION MATRIX:

CO Statement (BCS-DS-652)	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
BCS-DS-652.1	1	-	1	-	-	-	-	-	-	2	-	-	2	2	-
BCS-DS-652.2	3	-	2	2	1	-	-	-	1	2	2	-	-	1	2
BCS-DS-652.3	2	-	3	1	1	-	-	-	1	-	-	-	2	3	3
BCS-DS-652.4	3	1	3	1	-	-	-	-	-	-	-	2	2	2	2
BCS-DS-652.5	2	2	3	1	-	-	-	-	-	-	2	2	-	3	-
BCS-DS-652.6	3	3	3	2	3	-	-	1	2	3	1	2	3	-	3

BCS-DS-611: Deep Learning

Periods/week Credits		Max. Marks :200
L 3 T:0 P :0	3	Continuous Evaluation :100
Duration of Exam: 3 Hrs		End Term Examination :100

Pre-Requisite: Machine learning (BCS-DS-602)

Course Type: Program Core

Course Outcomes: The students will be able to-
BCS-DS-611.1. Understanding the basics concepts of deep learning.
BCS-DS-611.2. Emphasizing knowledge on various deep learning algorithms, which are more appropriate for various types of learning tasks in various domains.

BCS-DS-611.3. Understanding of CNN and RNN to model for real world applications.
BCS-DS-611.4. Understanding the various challenges involved in designing deep learning algorithms for varied applications

PART- A

Unit-1: INTRODUCTION

1.1. Introduction to Deep Learning: Basics: Biological Neuron, History of Deep Learning,
1.2. Idea of computational units, McCulloch– Pitts unit and Thresholding logic,
1.3. Linear Perceptron, Perceptron Learning Algorithm,
1.4. Linear separability, Convergence theorem for Perceptron Learning Algorithm

Unit-2: FEEDFORWARD NETWORKS

1.5. Feedforward Networks: Multilayer Perceptron,
1.6. Gradient Descent, Hidden Units, Architecture Design,
1.7. Backpropagation, Empirical Risk Minimization, regularization, auto encoders

Unit-3: CONVOLUTIONAL NETWORKS

3.6. The Convolution Operation, Motivation, Pooling, Convolution and Pooling as an Infinitely Strong Prior,
3.7. Variants of the Basic Convolution Function, Structured Outputs, Data Types,
3.8. Efficient Convolution Algorithms, Random or Unsupervised Features

PART –B

Unit-4: SEQUENCE MODELLING

4.1 Recurrent and Recursive Nets: Unfolding Computational Graphs,
4.2 Recurrent Neural Networks, Bidirectional RNNs,
4.3 Encoder-Decoder Sequence-to-Sequence Architectures,
4.4 Deep Recurrent Networks And Recursive Neural Networks. Long short-term memory

Unit-5:RECURRENT NEURAL NETWORKS

5.1 Recurrent Neural Networks: Bidirectional RNNs,
5.2 Deep Recurrent Networks Recursive Neural Networks,
5.3 The Long Short-Term Memory and Other Gated RNNs

Unit-6:DEEP GENERATIVE MODELS

6.1 Deep Generative Models: Boltzmann Machines,
6.2 Restricted Boltzmann Machines, Introduction to MCMC and Gibbs Sampling- gradient computations in RBMs,
6.3 Deep Belief Networks, Deep Boltzmann Machines

Text Books / Reference Books:

1. Ian Goodfellow, YoshuaBengio, Aaron Courville, "Deep Learning", MIT Press, 2016.
2. Bengio, Yoshua. "Learning deep architectures for AI." Foundations and trends in Machine Learning 2.1 (2009): 1127.
3. N.D.Lewis, "Deep Learning Made Easy with R: A Gentle Introduction for Data Science", January 2016.
4. Nikhil Buduma, "Fundamentals of Deep Learning: Designing Next-Generation Machine Intelligence Algorithms", O'Reilly publications.
5. Michael Nielsen, Neural Networks and Deep Learning, Determination Press, 2015.
6. CosmaRohillaShalizi, Advanced Data Analysis from an Elementary Point of View, 2015.
7. Deng & Yu, Deep Learning: Methods and Applications, Now Publishers, 2013.

Instructions for paper setting: Seven questions are to be set in total. First question will be conceptual covering entire syllabus and will be compulsory to attempt. Three questions will be set from each Part A and Part B (one from each unit) Student needs to attempt two questions out of three from each part. Each question will be of 20 marks.

Distribution of Continuous Evaluation:

Sessional- I	30%
Sessional- II	30%
Assignment/Tutorial	20%
Class Work/ Performance	10%
Attendance	10%

Evaluation Tools:

Assignment/Tutorials
Sessional tests
Surprise questions during lectures/Class Performance
Term end examination

COURSE ARTICULATION MATRIX :

CO Statement	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
BCS-DS-611.1	2	2	2	2		2		1	2		2	1		2	2
BCS-DS-611.2	2	2	1	1		1		2		2	1	1		1	
BCS-DS-611.3	1	1		2	1	2	1		2		1			2	
BCS-DS-611.4	1	1		1			1	1	2	2		2	2		2

MANAV RACHNA INTERNATIONAL INSTITUTE OF RESEARCH AND STUDIES
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NAAC 'A++' Grade University
BCS-DS-613: Emerging Computing Technologies

Periods/week	Credits	Max. Marks:	200
L :3 T: 0	2.0	Continuous Evaluation	: 100
Duration of Exam: 2 Hrs		End Sem Evaluation	: 100

Pre-Requisite: NIL
Course Type: Program Core

Course Outcomes: Students will be able to:
BCS-DS-613.1. to understandagile methodology and full stack environment for software development
BCS-DS-613.2. to apply security tactics and patterns to enhance the security of software applications
BCS-DS-613.3. to analyse concepts of DevOps and Cloud Computing and its evolution from traditional IT practices
BCS-DS-613.4. to develop software solution with industry standard development method.

Unit 1- Agile Methodology
1.1 Introduction, Agile vs. Traditional Project Development (Waterfall Model, Iterative Enhancement Model), Agile Mindset
1.2 Project Life Cycles, Implementation Preparation
1.3 Introduction to Scrum, Kanban andAdditional Agile Approaches
1.4 Measuring Agile Projects, Ensuring Success

Unit 2 Full Stack Development
2.1 Front-end Development: HTML/CSS, JavaScript, Frameworks like React, Angular, or Vue.js, Responsive web design, Cross-browser compatibility
2.2 Back-end Development: Server-side languages like Node.js, Python, or Ruby, API design and integration, Database management (SQL or NoSQL), Server management and deployment
2.3 Version Control/Git: Proficiency in using Git for source code management and collaboration.
2.4 Databases: Understanding of databases and data modeling is crucial for creating efficient, scalable applications.
2.5 Web Architecture: Knowledge of web architecture and how various components interact.

Unit 3: Cloud Computing
3.1 Cloud computing - Introduction
3.2 Cloud Foundations: Introduction to Virtualization (VM's and Containers), Service Delivery & Deployment Models
3.3 Azure Compute Infrastructure: Infrastructure Monitoring, Storage, Virtual Machines, Deployment & Configuration, Virtual Networking, Azure Active Directory, Azure App Solutions, Implement Application Infrastructure, Implement Container based applications
3.4 Azure Data: NoSQL databases, SQL databases and related security

Unit 4Software Security

4.1 Significance of software security: Software security risk management, Software security resources
4.2 Software Security Threats: Threats to software security, Hardware-level, Code-level, Detailed design-level, Architectural-level, Requirements-level, Threat modeling and tools
4.3 Secure Software Design: Introduction to secure design, Security tactics, Security patterns, Security vulnerabilities, Architectural analysis for security,
4.4 Case studies:Pattern-Oriented Architectural Analysis, Vulnerability-Oriented Architectural Analysis, Software security anti-patterns
4.5 Recent Development and Future Directions: Developer-friendly software security, IoT and software security

Unit 5 Trends in DevOps
5.1 Overview of DevOps, and Tradition IT, DevOps vs Agile
5.2 GitOps: Benefits, Applications and PopularPlatforms
5.3 MLOps: benefits, Applications and popular platforms
5.4 AIOps: benefits, Applications and popular platforms
5.5 Data Ops: Frameworks/Tools and best practices
5.6 DevSecOps: Benefits, Applications, Framework/Tools, best practices
5.7 DevOps: Case Studies and Real-world Implementations

Software required/Weblinks:
www.w3schools.com

Instructions for paper setting: MCQ based evaluation. Equal weightage will be given to each unit
Distribution of Continuous Evaluation:

Sessional-I	30%
Sessional- II	30%
Assignment	20%
Class Performance	10%
Attendance	10%

Evaluation Tools:
Sessional-I
Sessional-II
Class Performance
Assignment
Attendance
Course Articulation Matrix:

CO Statement (BCS-DS-613)	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
BCS-DS-613.1	-	-	-	-	-	-	1	1	2	-	-	-	-	2	2
BCS-DS-613.2	-	2	-	-	2	3	1	2	-	-	-	-	-	2	2
BCS-DS-613.3	-	-	-	-	-	-	-	-	2	-	-	-	-	2	2
BCS-DS-613.4	-	1	2	-	-	-	-	-	2	-	-	-	-	3	2

BCS-DS-659: Deep Learning Lab

Periods/week	Credits	Max. Marks	:100
P :2	1	Continuous Evaluation	:50
Duration of Exam: 2 Hrs		End Term Examination	: 50

Pre-Requisite: Machine learning Lab- (BCS-DS-652)
Course Type: Program Core

Course Outcomes: The students will be able to-
BCS-DS-659.1 Preprocess the dataset for building a model.
BCS-DS-659.2 Implement various deep learning techniques suitable for a given problem.
BCS-DS-659.3 Gain understanding of Natural Language Processing, recommender systems, and other types of classifiers.
BCS-DS-659.4 Build, train, and deploy real world applications such as object recognition and Computer Vision.

NOTE: The laboratory should be preceded or followed by a tutorial to explain the approach or algorithm to be implemented for the problem given.

List of Practicals:

1 Introduction: Get your first taste of deep learning by applying style transfer to your own images, and gain experience using development tools such as Anaconda and Jupyter notebooks.

2 Neural Networks: Learn neural networks basics, and build your first network with Python and NumPy. Use the modern deep learning framework PyTorch to build multi-layer neural networks, and analyze real data.
3 Convolutional Neural Networks: Learn how to build convolutional networks and use them to classify images (faces, melanomas, etc.) based on patterns and objects that appear in them. Use these networks to learn data compression and image denoising.
4 Recurrent Neural Networks: Build your own recurrent networks and long short-term memory networks with PyTorch; perform sentiment analysis and use recurrent networks to generate new text from TV scripts.
5 Generative Adversarial Networks: Learn to understand and implement the DCGAN model to simulate realistic images, with Ian Goodfellow, the inventor of GANS (generative adversarial networks).
6 Deploying a Sentiment Analysis Model: Use deep neural networks to design agents that can learn to take actions in a simulated environment. Apply reinforcement learning to complex control tasks like video games and robotics.
7 SAMPLE PROJECT 1 Predicting Bike-Sharing Patterns Build and train neural networks from scratch to predict the number of bikeshare users on a given day.
8 SAMPLE PROJECT 2 Dog-Breed Classifier Design and train a convolutional neural network to analyze images of dogs and correctly identify their breeds. Use transfer learning and well-known architectures to improve this model—this is excellent preparation for more advanced applications.
9 SAMPLE PROJECT 3 Generate TV scripts Build a recurrent neural network on TensorFlow to process text. Use it to generate new episodes of your favorite TV show, based on old scripts.
10 SAMPLE PROJECT 4 Generate Faces Build a pair of multi-layer neural networks and make them compete against each other in order to generate new, realistic faces. Try training them on a set of celebrity faces, and see what new faces the computer comes out with!
11 SAMPLE PROJECT 5 Deploying a Sentiment Analysis Model Train and deploy your own PyTorch sentiment analysis model. You'll build a model and create a gateway for accessing it from a website.

Softtware required/Weblinks:
www.tutorialpoint.com
www.nptel.com

Reference Books/Online Resources:

1. Deep Learning with Python by Francois Chollet
2. Practical Convolutional neural network by MohitSewak, Md. Rezaul Karim and Pradeep Pujari

Note: At least 5 more exercises to be given by the teacher concerned.

Distribution of Continuous Evaluation:

Viva- I	30%
Viva- II	30%
File/Records	20%
Class Work/ Performance	10%
Attendance	10%

Evaluation Tools:
Experiments in lab

File work/Class Performance
Viva (Question and answers in lab)
End Term Practical Exam

COURSE ARTICULATION MATRIX :

CO Statement	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO 1**	**PSO 2**	**PSO 3**
BCS-DS--659.1	1	2	1	1	2	1	1	1	-	2	-	-	2	2	2
BCS-DS-659.2	1	1		1	-	1	-	2	2	2	1	2	-	-	-
BCS-DS-659.3	2		1	1	1	-	2	-	-	-	1	2	2	2	-
BCS-DS-659.4	1	1	1		2	1	1	2	2	2	1	-	2	1	2

BCS-DS-802: Reinforcement Learning

Periods/week	Credits	Max. Marks	:200
L : 3 T: 0	3. 0	Continuous Evaluation	:100
Duration of Exam: 3Hrs		End Term Examination	: 100

Pre-Requisite: Machine Learning(BCS-DS-602)
Course Type: Programme Core

Course Outcomes: The students will be able to-

BCS-DS-802.1. Knowledge of basic and advanced reinforcement learning techniques
BCS-DS-802.2. Identification of suitable learning tasks to which these learning techniques can be applied.
BCS-DS-802.3. Formulation of decision problems, set up and run computational experiments, evaluation of results from experiments.
BCS-DS-802.4. Evaluate empirical performance through various algorithm.

PART- A

Unit-1: Introduction

1.1 Origin and history of Reinforcement Learning research.
1.2 Its connections with other related fields and with different branches of machine learning.
1.3 Probability concepts - Axioms of probability, concepts of random variables, PMF, PDFs, CDFs, Expectation.
1.4 Concepts of joint and multiple random variables, joint, conditional and marginal distributions. Correlation and independence

Unit-2:Markov Decision Process

2.1 Introduction to RL terminology, Markov property, Markov chains, Markov reward process (MRP).
2.2 Introduction to and proof of Bellman equations for MRPs along with proof of existence of solution to Bellman equations in MRP.

2.3 Introduction to Markov decision process (MDP), state and action value functions, Bellman expectation equations, optimality of value functions and policies, Bellman optimality equations.

Unit-3: Prediction and Control by Dynamic Programming

3.1 Overview of dynamic programming for MDP, definition and formulation of planning in MDPs, 3.2 Principle of optimality, iterative policy evaluation, policy iteration, value iteration, Banach fixed

point theorem, proof of contraction mapping property of Bellman expectation and optimality operators,

3.3 proof of convergence of policy evalutation and value iteration algorithms, DP extension

PART –B

Unit-4: Monte Carlo Methods and TD Methods

4.1 Overiew of Monte Carlo methods for model free RL, First visit and every visit Monte Carlo,

4.2 Monte Carlo control, On policy and off policy learning, Importance sampling.Incremental Monte Carlo Methods for Model Free Prediction,

4.3 Overview TD(0), TD(1) and TD(λ), k-step estimators, unified view of DP.

Unit-5:Function Approximation Method and Policy Gradients

5.1 Introduction to function approximation methods, Revisiting risk minimization, gradient descent from Machine Learning,

5.2 Gradient MC and Semi-gradient TD(0) algorithms, Eligibility trace for function approximation, Afterstates,Control with function approximation, Least squares,

5.3 Experience replay in deep Q-Networks.policy gradient methods, Log-derivative trick,

5.4 Naive REINFORCE algorithm, Reducing variance in policy gradient estimates

Text Books / Reference Books:

1. Reinforcement Learning: An Introduction, Sutton and Barto, 2nd Edition.
2. Richard S. Sutton and Andrew G. Barto, "Reinforcement learning: An introduction", Second Edition, MIT Press, 2019
3. Li, Yuxi. "Deep reinforcement learning." arXiv preprint arXiv:1810.06339 (2018).
4. Wiering, Marco, and Martijn Van Otterlo. "Reinforcement learning." Adaptation, learning, and optimization 12 (2012)
5. Russell, Stuart J., and Peter Norvig. "Artificial intelligence: a modern approach."Pearson Education Limited, 2016.
6. Goodfellow, Ian, YoshuaBengio, and Aaron Courville. "Deep learning." MIT press, 2016..

Software required/Weblinks:

1. https://www.researchgate.net/topic/Reinforcement-Learningwww.nptel.com

Instructions for paper setting: Seven questions are to be set in total. First question will be conceptual covering entire syllabus and will be compulsory to attempt. Three questions will be set from each Part A and Part B (one from each unit) Student needs to attempt two questions out of three from each part. Each question will be of 20 marks.

Distribution of Continuous Evaluation:

Sessional- I	30%
Sessional- II	30%

Assignment/Tutorial	20%
Class Work/ Performance	20%
Attendance	10%

Evaluation Tools:
Assignment/Tutorials
Sessional tests
Surprise questions during lectures/Class Performance
Term end examination

COURSE ARTICULATION MATRIX :

CO Statement	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
BCS-DS-802.1	3	3	3	2	2	2	-	2	1	2	2	2	2	2	2
BCS-DS-802.2	3	3	3	3	2	2	-	2	1	2	2	2	2	-	2
BCS-DS-802.3	3	3	3	-	3	2	1	-	3	3	-	1	2	3	2
BCS-DS-802.4	3	2	2	2	3	2	2	-	2	3	-	3	2	3	2

BCS-DS-851: Reinforcement Learning Lab

Periods/week	Credits	Max. Marks	:100
P :2	1	Continuous Evaluation	:50
Duration of Exam: 2 Hrs		End Term Examination	: 50

Pre-Requisite: Machine Learning Lab(BCS-DS-652)
Course Type: Programme Core

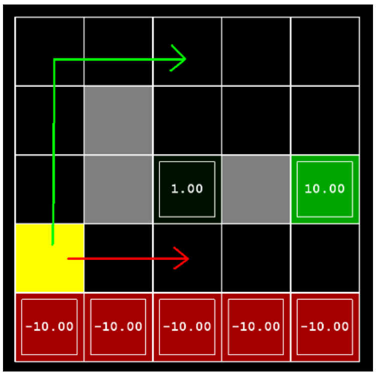
Course Outcomes: The students will be able to-
BCS-DS-851.1 Understand the implementation procedures for the reinforcement learning algorithms.
BCS-DS-851.2 Apply appropriate data sets to the algorithms.
BCS-DS-851.3 Identify different reinforcement algorithms to solve real world problems
BCS-DS-851.4 Formulate clustering algorithm for solving the problem.

NOTE: The laboratory should be preceded or followed by a tutorial to explain the approach or algorithm to be implemented for the problem given.

List of Practicals:

1 Write a value iteration agent in ValueIterationAgent, which has been partially specified for you in valueIterationAgents.py. Your value iteration agent is an offline planner, not a reinforcement agent, and so the relevant training option is the number of iterations of value iteration it should run (option -i) in its initial planning phase..

2. Consider the DiscountGrid layout, shown below. This grid has two terminal states with positive payoff (shown in green), a close exit with payoff +1 and a distant exit with payoff +10. The bottom row of the grid consists of terminal states with negative payoff (shown in red); each state in this "cliff" region has payoff -10. The starting state is the yellow square. We distinguish between two types of paths: (1) paths that "risk the cliff" and travel near the bottom row of the grid; these paths are shorter but risk earning a large negative payoff, and are represented by the red arrow in the figure below. (2) paths that "avoid the cliff" and travel along the top edge of the grid. These paths are longer but are less likely to incur huge negative payoffs. These paths are represented by the green arrow in the figure below.



3 Implement Monte Carlo control off – policy control using Weighted Importance Sampling

4 Write a q-learning agent by implementing epsilon-greedy action selection in get Action, meaning it chooses random actions epsilon of the time, and follows its current best q-values otherwise

5 Implement an approximate q-learning agent that learns weights for features of states, where many states might share the same features. Write your implementation in Approximate QAgent class in qlearningAgents.py, which is a subclass of PacmanQAgent.

6 Implement Q Learning Algorithm to play and design Pac man Game

7 Create an epsilon – greedy policy based on a given Q-function and epsilon.Find the optimal greedy policy while following an epsilon –greedy policy

Software required/Weblinks:

1. https://www.researchgate.net/topic/Reinforcement-Learningwww.nptel.com

Text Books:

1) Reinforcement Learning: An Introduction, Sutton and Barto, 2nd Edition.
2) Richard S. Sutton and Andrew G. Barto, "Reinforcement learning: An introduction", Second Edition, MIT Press, 2019
3) Li, Yuxi. "Deep reinforcement learning." arXiv preprint arXiv:1810.06339 (2018).
4) Wiering, Marco, and Martijn Van Otterlo. "Reinforcement learning." Adaptation, learning, and optimization 12 (2012)
5) Russell, Stuart J., and Peter Norvig. "Artificial intelligence: a modern approach."Pearson Education Limited, 2016.
6) Goodfellow, Ian, YoshuaBengio, and Aaron Courville. "Deep learning." MIT press, 2016..

Note: At least 5 more exercises to be given by the teacher concerned.

Distribution of Continuous Evaluation:

Viva- I	30%
Viva- II	30%
File/Records	20%
Class Work/ Performance	10%
Attendance	10%

Evaluation Tools:

Experiments in lab

File work/Class Performance

Viva (Question and answers in lab)

End Term Practical Exam

COURSE ARTICULATION MATRIX :

CO Statement	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
BCS-DS-851.1	3	3	3	2	-	-	-	-	-	-	-	2	2	2	2
BCS-DS-851.2	3	2	2	2	-	-	-	-	-	-	-	3	2	2	2
BCS-DS-851.3	3	3	3	1	-	-	-	-	-	-	-	2	2	3	2
BCS-DS-851.4	3	3	3	3	-	-	-	-	-	-	-	2	2	2	2

BCS-DS-635: Internet of Things for AI

Periods/week Credits	Max. Marks :200
L :2 T: 0 2.0	Continuous Evaluation : 100
Duration of Exam: 3 Hrs	End Term Examination : 100

Pre-Requisite: NIL
Course Type: Programme Elective
Course Outcomes: The students will be able to-
BCS-DS-635.1. Describe concepts and architecture of IoT systems
BCS-DS-635.2 Explain the key technologies and protocols employed at each layer of the IoT stack.
BCS-DS-635.3 Identify different infrastructure components including sensors, embedded hardware, and network devices for IoT applications.
BCS-DS-635.4 Apply the knowledge and skills acquired during the course to build and test a complete, working IoT system involving prototyping, programming and data analysis.

PART- A

Unit-1: Introduction to IoT
1.1 Introduction to IoT Communications, layered architectural overview of IoT,
1.2 IoT components and implementation: Functional blocks, Sensing,
1.3 Actuation, Intelligent Analytics, Machine to Machine Communication,
1.4 IoT vs M2M, IoT Enabling Technologies. IoT use cases.

Unit-2: IoT Communication Fundamentals
2.1 IoT Network Configurations: Node, PAN, LAN, WAN,
2.2 Gateway, Proxy; IoT Communication: IoT Protocol Stack,
2.3 IoT Communication Models,
2.4 Connectivity technologies: IEEE802.15.4, ZigBee, 6LOWPAN,
2.5 Bluetooth, Z-wave, RFID, MQTT, CoAP, XMPP, AMQP

Unit-3: Sensor and Cloud Networks

3.1 RF Wireless Sensors, Wireless Sensor Networks, Components of a sensor node, Challenges in Wireless sensor networks, Node Cooperation,

3.2 WSN Coverage, Network Devices in IoT systems, Cloud computing in IoT, Services (Infrastructure, Software, Platform),

3.3 Device Data Storage on Cloud: Sensor Cloud Architecture and Applications, Fog Computing: Requirement and Architecture

PART –B

Unit-4: IoT Implementation for Application Development

4.1 Component based IoT Reference Model, Introduction to Arduino,

4.2 Raspberry Pi boards, Arduino and Python Programming for IoT applications development,

4.3 Role of Artificial Intelligence in Internet of Things, IoT Data Handling and Management,

4.4 IoT Data Analytics

Unit-5: IoT Security Aspects and Use Cases

5.1 Security and Privacy Aspects in IoT, Principles of Secure IoT Communication,

5.2 IoT Security Framework. AI enabled IoT case studies: Home Automation,

5.3 Asset Management, Tagging and Tracking for Healthcare Applications,

5.4 Smart Irrigation, Connected Vehicles, Industrial IoT, Smart Cities.

Text Books / Reference Books:

1. Vijay Madisetti, ArshdeepBahga, İnternet of Things, "A Hands-on Approach", University Press
2. Jeeva Jose, "Internet of Things", Khanna Publishing House, Delhi
3. Adrian McEwen, "Designing the Internet of Things", Wiley
4. Raj Kamal, "Internet of Things: Architecture and Design", McGraw Hill
5. Dr. SRN Reddy, RachitThukral and Manasi Mishra, "Introduction to Internet of Things: A practical Approach", ETI Labs
6. Pethuru Raj and Anupama C. Raman, "The Internet of Things: Enabling Technologies, Platforms, and Use Cases", CRC Press
7. CunoPfister, "Getting Started with the Internet of Things", O Reilly Media

Software required/Weblinks:

1. https://onlinecourses.nptel.ac.in/noc17_cs22
2. www.analyticsvidya.com

Distribution of Continuous Evaluation:

Sessional- I	30%
Sessional- II	30%
Assignment/Tutorial	20%
Class Work/ Performance	10%
Attendance	10%

Evaluation Tools:

Assignment/Tutorials
Sessional tests
Surprise questions during lectures/Class Performance
Term end examination

COURSE ARTICULATION MATRIX :

CO Statement	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
BCS-DS-635.1	3	2	1	2	2	2	1	1	2	1	1	3	2	1	3
BCS-DS-635.**.2**	3	3	3	3	2	2	2	2	2	1	1	3	1	2	3
BCS-DS-635.**3**	3	3	3	3	3	2	2	2	2	3	3	3	3	3	3
BCS-DS-635.4	3	3	3	3	3	2	2	2	2	3	3	3	3	3	3

BCS-DS-684:Internet of Things for AI Lab

Periods/week	Credits	Max. Marks	:100
P :2	1	Continuous Evaluation	:50
Duration of Exam: 2 Hrs		End Term Examination	: 50

Pre-Requisite: NIL
Course Type: Programme Elective
Course Outcomes: The students will be able to-
BCS-DS-684.1 To identify different infrastructure components including sensors, embedded hardware, gateways and network systems for specified IoT application.
BCS-DS-684.2 To set up an embedded/microcomputer system by interface I/O devices, sensors and enable to develop IoT application on it.
BCS-DS-684.3 To establish a secure and consistent communication from microcontroller/micro computer system to the cloud through communication modules.
BCS-DS-684.4 To design an application to create communication with multiple sensors to store data locally and remotely monitor data and control devices.
BCS-DS-684.5 To apply the knowledge and skills acquired during the course to design IoT based solutions for real world problems...

NOTE: The laboratory should be preceded or followed by a tutorial to explain the approach or algorithm to be implemented for the problem given.

List of Practicals:
1. Familiarization with Arduino board and perform necessary software installation.
2. To interface LED with Arduino and write a program to blink LED at defined delay.
3. To interface Push button/Digital sensor (IR/LDR) with Arduino and write a program to turn ON LED when push button is pressed or at sensor detection.
4. Working with A/D conversion and sensor integration.
5. To interface Bluetooth with Arduinoand write a program to turn LED ON/OFF when ‘A’/’B’ is received from smart phone using Bluetooth.
6. Creating own Android App using MIT App Inventor, controlling Arduino connected devices and sending data to cloud.
7. Introduction to Raspberry-Pi: Setup and Procedure.
8. Controlling GPIO Pins of Raspberry-Pi to blink interfaced LED.
9. Sending sensor data to cloud via R-Pi.

10. To interface DHT11 sensor with Arduino/Raspberry Pi and write a program to upload temperature and humidity readings to cloud.
11. Write a program on Arduino/Raspberry Pi to publish temperature data to MQTT broker.
12. Write a program on Arduino/Raspberry Pi to subscribe to MQTT broker for temperature data and print it.

Softtware required/Weblinks:

https://www.arduino.cc/
https://www.raspberrypi.org/

Text Books:

1. Dr. SRN Reddy, RachitThukral and Manasi Mishra, "Introduction to Internet of Things: A practical
   Approach", ETI Labs
2. Jeeva Jose, "Internet of Things", Khanna Publishing House, Delhi
3. Adrian McEwen, "Designing the Internet of Things", Wiley
4. CunoPfister, "Getting Started with the Internet of Things", O Reilly Media

Note: At least 5 more exercises to be given by the teacher concerned.

Distribution of Continuous Evaluation:

Viva- I	30%
Viva- II	30%
File/Records	20%
Class Work/ Performance	10%
Attendance	10%

Evaluation Tools:

Experiments in lab
File work/Class Performance
Viva (Question and answers in lab)
End Term Practical Exam

COURSE ARTICULATION MATRIX :

CO Statement	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO 1**	**PSO 2**	**PSO 3**
BCS-DS-684.1	1	3	-	-	-	-	-	-	-	-	-	1	-	-	-
BCS-DS-684.2	-	-	3	2	-	-	-	-	-	-	-	1	-	-	-
BCS-DS-684.3	-	-	-	-	3	2	-	-	-	-	-	1	-	-	-
BCS-DS-684.4				1	3	3						1			
BCS-DS-684.5					2		3		1		2	1			

MANAV RACHNA INTERNATIONAL INSTITUTE OF RESEARCH AND STUDIES
(Deemed to be University under section 3 of the UGC Act 1956)
NAAC 'A++' Grade University

HM-607: GERMAN – II

Periods/week Credits	Max. Marks	: 100
L: 2 T: 0 2.0	Continuous Evaluation	: 50
Duration of Examination: 1.5 Hrs	End Sem Examination	: 50

Pre-Requisites: NIL
Course Type: Generic Elective

Course Outcomes: Students will be able to-
HM-607.1. Discuss about various directions, countries and languages they speak.
HM-607.2. Write short essays on family and friends. They will have knowledge of tenses.
HM-607.3. Identify classroom vocabulary in the German language.
HM-607.4. Speak ordinal and cardinal numbers and they will also learn months, days in German.
HM-607.5. Express or/and justify opinions using equivalents of different verbs.
HM-607.6. Describe themselves, other people, familiar places and objects in short discourse using simple sentences and basic vocabulary.

PART – A

Unit 1: Ordinal und Kardinal Zahlen
1.1 Ordinal & Cardinal numbers
1.2 Months, days, Feiertage and dates.

Unit 2: sein und haben
2.1 Verbs: to be and to have
2.2 helping verbs practice worksheets
2.3 Vocabulary (Family) short essay on family, friends etc.

PART – B

Unit 3: Gegenstände im Kursraum
3.1 Vocabulary (classroom)
3.2 Definite and indefinite articles

Unit 4: Länder, Sprachen
4.1 Countries, languages, directions
4.2 Past of the verb 'to be'

Text Books/Reference Books:

1. Rita Maria Niemann,Cornelsen, 2005, Studio d A1: Deutsch alsFremdsprache, Volume 6.
2. Dallapiazza, Rosa-Maria and Jan, Eduard von. Tangram aktuell 1. Deutsch alsFremdsprache Tangram aktuell 1 - Lektion 1-4: Deutsch als. (HueberVerlag, 2005).
3. Dallapiazza, Rosa-Maria and Jan, Eduard von. Tangram aktuell 1. Deutsch alsFremdsprache Tangram aktuell 1 - Lektion 5-8: Deutsch als. (HueberVerlag, 2005).

4. Paul Rusch, 2015: Langenscheidt and Klett.

Weblinks:
http://www.nthuleen.com/

Instructions for paper setting: Seven questions are to be set in total. First question will be conceptual covering entire syllabus and will be compulsory to attempt. Student needs to attempt four questions from the remaining six questions. Five questions need to be attempted in total. Each question will be of 10 marks.

Evaluation Tools:

Sessional- I	30%
Sessional- II	30%
Assignment/Tutorial	20%
Class Work/ Performance	10%
Attendance	10%

COURSE ARTICULATION MATRIX :

CO Statement (HM–607)	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
HM-607.1	-	-	-	-	-	1	-	-	1	1	-	1	-	-	1
HM-607.2	-	-	-	-	-	1	-	-	1	1	-	-	-	-	1
HM-607.3	-	-	-	-	-	1	-	-	1	1	-	-	-	-	1
HM-607.4	-	-	-	-	-	1	-	-	1	1	-	-	-	-	1
HM-607.5	-	-	-	-	-	1	-	-	1	1	-	1	-	-	1
HM-607.6	-	-	-	-	-	1	-	-	1	1	-	-	-	-	1